Reg. No. | | | | | | | | | | | [17AE42]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE-2019

AEROSPACE VEHICLES AND SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed.

PART - A

1. a. Write a short note on the contributions of the Wright Brothers to the development of aeronautical engineering. **04**

b. What is meant by a swing-wing/ variable sweep aircraft? What are the advantages and disadvantages of this aircraft design? **05**

c. Define the International Standard Atmosphere. Describe temperature varies with altitude in the ISA. **05**

2. a. Describe the different types of high lift devices that are used on fixed wing aircraft. Provide a neat sketch and draw airflow patterns for different types of flaps commonly used on aircraft. **06**

b. With neat sketches describe collective and cyclic control of a conventional helicopter. **08**

3. a. Compare and contrast the advantages and disadvantages of the aerospace applications of the Brayton and OTTO cycles. **07**

b. Explain the advantages and disadvantages of composite materials used in the aerospace industry. **07**

4. a. **i.** Explain the concepts of rigidity and precession in a gyroscope. **03**

ii. Explain how gyroscopic properties are used in the attitude indicator/artificial horizon. **04**

b. Write short notes on:

i. Distance Measuring Equipment (DME) **03**

ii. VHF Omni range (VOR) System. **04**

PART - B

5. a. Explain the following:

i. Aerial Refuelling: **04**

ii. Alternator/Generator **03**

b. **i.** Draw a neat diagram of a basic radio transmitter circuit showing its components and explain the function of each component in brief. **04**

ii. Draw a neat diagram of a basic radio receiver circuit showing its components and explain the function of each component in brief. **03**

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6. a. Draw the flight envelope/V-n diagram for an aircraft and explain its significance for flight. **04**

b. Plot the basic V-n diagram for the following acrobatic aircraft. Then, determine the maximum load factor. m = 2,300 kg, S = 19.33 m^2, C_{Lmax} = 2, - C_{Lmax} = -1.2, V_c = 600 kmph . Given: ρ = 1.225kg/m^3, V_D = 1.55V_C, (Positive) n_{max} = +6; (Negative) n_{max} = - (0.0.5 x 6) = -3. **10**

7. a. Define the following in an orbit: **3x2**

i. Eccentricity

ii. Semi major axis

iii. Inclination

b. State and explain the Kepler's laws of space dynamics with neat diagrams. **08**

8. a. Write short notes (with neat diagrams where appropriate) on:

i. Grain configurations in solid propellant rockets. **04**

ii. How specific impulse can be used to measure the effectiveness of a rocket motor. **03**

iii. Advantages and disadvantages of nuclear rockets. **03**

b. Compare and contrast strategic missiles with tactical missiles. **04**